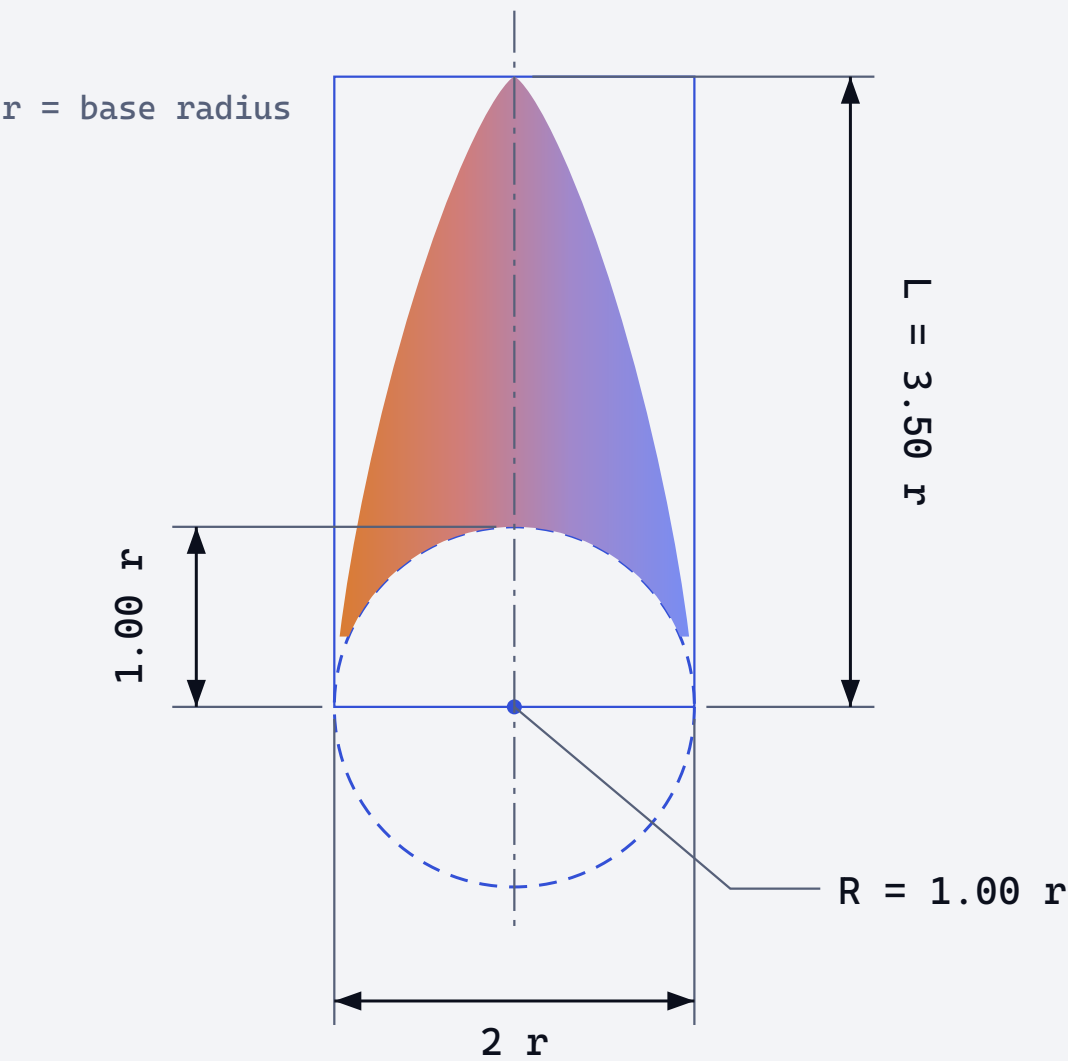
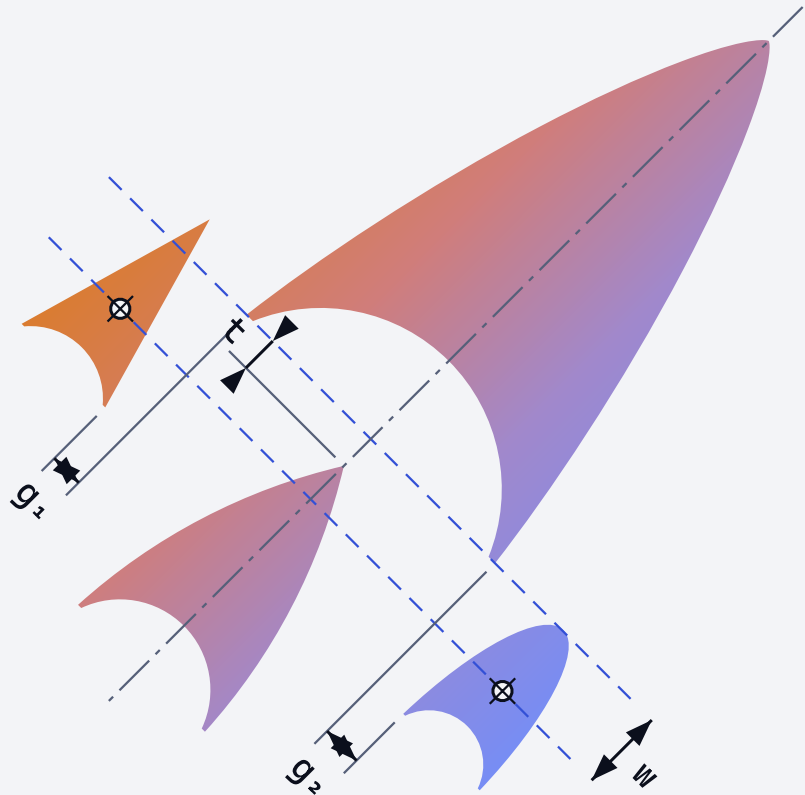


1 · CONE (VON KÁRMÁN)

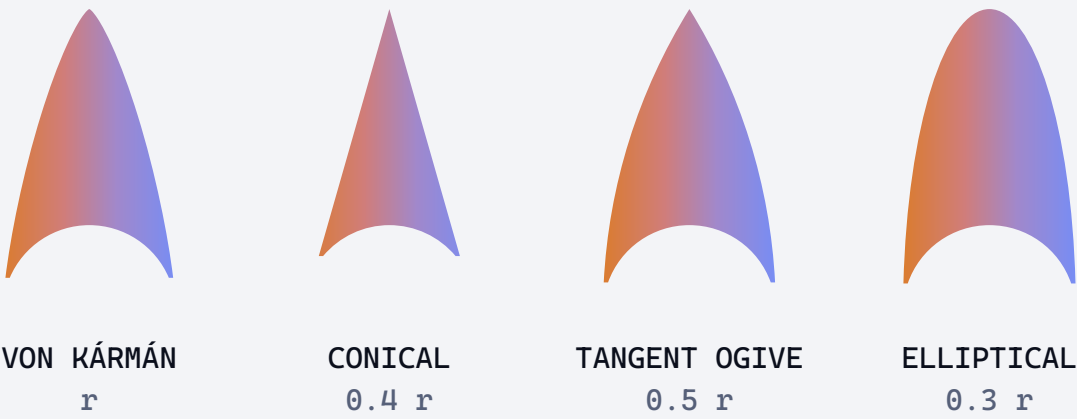


2 · SPACING ON THE 45° GRID

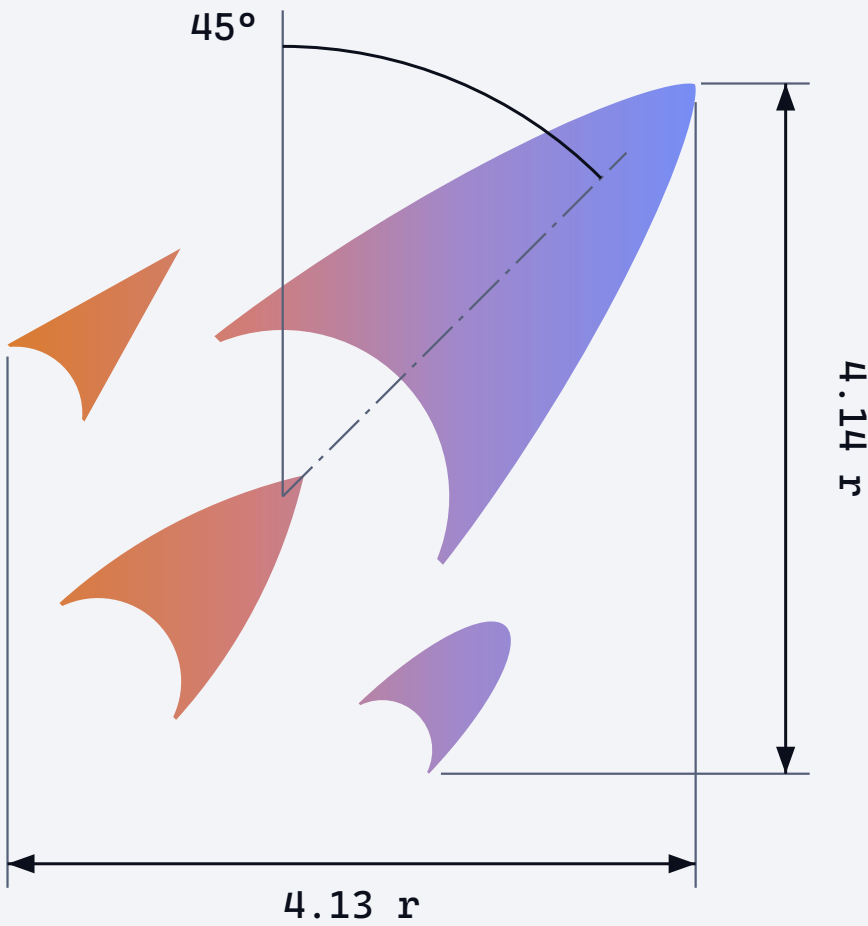


$g_1 = 0.19 r$ ,  $g_2 = 0.23 r$  from the Von Kármán foot corners to the wings. Wing centroids share a line  $w = 0.47 r$  below the Von Kármán foot line. The ogive is the tail, on the same axis, tip  $t = 0.21 r$  below the Von Kármán foot line. Lay out on the grid, then the grid leans as one piece.

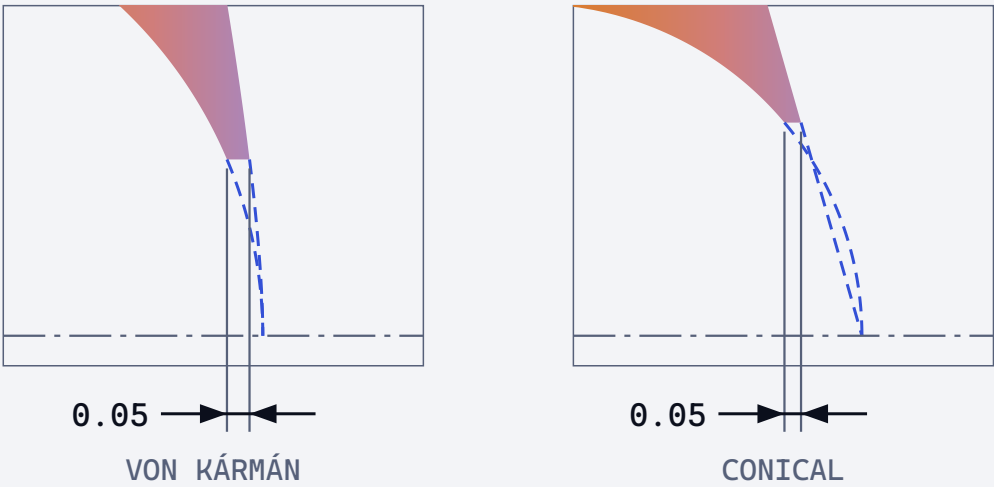
4 · PROFILES (EQUAL L / R)



3 · CLUSTER



5 · FEET (DETAIL)



Each foot is cut flat, parallel to the base (dash-dot), where the wall between flank and notch is  $0.05 \times$  the cone's base radius wide. Dashed: the uncut outline. On the conical cone the notch arc runs outside the straight flank near the base, so the cut also removes that sliver.

| CONE  | PROFILE       | RADIUS  | BASE CENTRE (u, v) |
|-------|---------------|---------|--------------------|
| MAIN  | VON KÁRMÁN    | $r$     | (+0.0, +0.0)       |
| WEST  | CONICAL       | $0.4 r$ | (-1.5, +0.8)       |
| EAST  | ELLIPTICAL    | $0.3 r$ | (+1.5, +0.7)       |
| SOUTH | TANGENT OGIVE | $0.5 r$ | (+0.0, +1.6)       |

u, v in  $r$  on the leaned grid, v down.

Standard nose-cone profiles,  $L = 3.50 \times$  base radius, notch  $1.00$  deep, feet cut flat where the wall is  $0.05$  wide (all  $\times$  base radius). Von Kármán is the LD-Haack series ( $C = 0$ ), drawn as smooth curves. The whole cluster leans  $45^\circ$  as one piece. One gradient spans the whole cluster, M orange left to O blue right.